



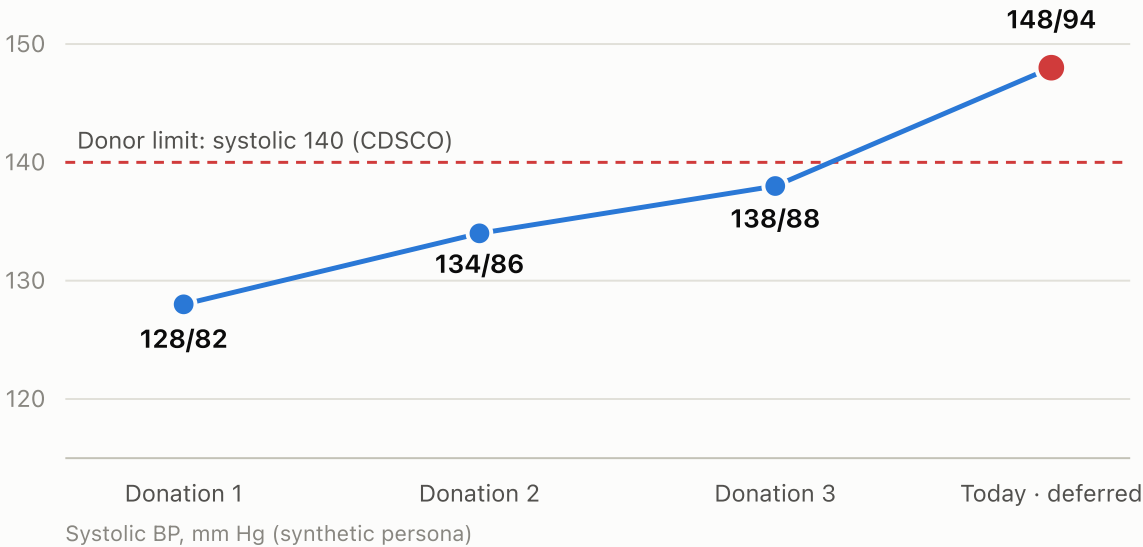
AI BUILDER CUP 2026 · SUSTAINABILITY & SOCIAL IMPACT

TraceDrop

Every blood donation is a health check.
TraceDrop makes sure it counts.

A Gemini-powered navigator that follows every finding from a blood donation through to care, with the blood centre's doctor and counsellor in charge.

"BP 148/94. You can't donate today. Come back later."



Arjun, 34 · software engineer, Bengaluru

- Gives blood at office camps. Never sees a doctor.
- The centre measured his BP at **every donation**. It rose each time, and **nobody joined the dots**.
- No symptoms, so he forgets. His father had a stroke at 58.

29%

of Indian adults aged 30–69 with raised BP **know they have it**.

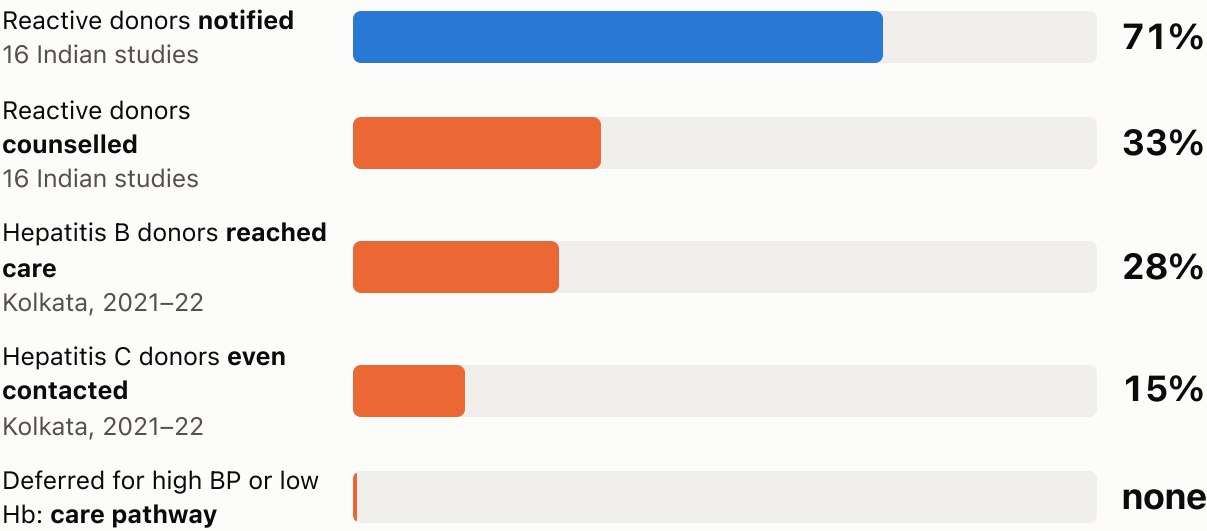
ICMR-NCDIR National NCD Monitoring Survey

THE PROBLEM

Blood donation finds disease in healthy adults, then loses them

India's **1.57 crore** donations a year make up its largest routine health check of working-age adults. Every donor's BP and haemoglobin are measured, and every unit is screened for infections.

What happens after a finding



Vox Sang 2024 (review of 16 studies); Kolkata blood centre study 2024. Hepatitis C is curable, and treatment is free under NVHCP.

0 / 235

letters to reactive donors in Delhi got a reply.

AJTS 2015

3.9 lakh

units discarded as reactive in 2025. Each is a person someone needs to reach.

Rajya Sabha Q1882 annex, our sum

35.5%

of Indian adults have hypertension (ICMR-INDIAB). Donors come from this population.

Results alone don't work. Follow-up does, and it needs a person.

Japan, since 1982

Results returned

The Red Cross sends donors 15 blood values through the Love Blood app.

Reference ranges are "not normal/abnormal judgements", and there's no next step.

Taiwan, 2025

Screening added

HbA1c and cholesterol tests for donors over 40.

42.7% didn't know they'd been screened. The study found "limited benefits... without additional interventions".

India, 2026

AI explainers exist

Aarogya Setu 2.0 "Smart Reports", AI trends, 92 crore ABHA IDs.

Plenty of records and trends, but nobody acts on them. Only 491 labs are connected to ABDM, against 27,283 hospitals.

The missing link is a navigator.

It has to know your history, explain why a finding matters to **you** in your language, book the next step and keep following up until it's done. That's what executives get from a physician and a care coordinator.

Every other piece already exists and much of it is free: Ayushman Arogya Mandir health centres, **eSanjeevani** (50 crore teleconsultations), free hepatitis treatment under **NVHCP**, and ABHA records. Nothing carries the donor to them.

Following every finding up by hand

~2,000

full-time staff a year (about 40 lakh hours)

~160

with a navigator doing the legwork, where people only disclose and approve

Estimate: 3.9 lakh reactive screens × 2 h, 12 lakh deferrals × 30 min, and a 10-minute review of every donation. Full assumptions are in the brief.

A navigator for every donor, with the blood centre in charge

Gemini · multimodal

1. Capture

Reads a register page, a lab-report photo or a PDF from any lab and writes one standard FHIR record. No integration needed.

Gemini · long context

2. Understand

Reasons over the donor's whole history using ICMR protocols. Safety thresholds stay as fixed rules.

Gemini · Indian languages

3. Explain

Explains why it matters to *them*, in their language, and answers "but I have no symptoms" with their own trend.

ADK agent · tools

4. Act

Books free care at an Ayushman Arogya Mandir, on eSanjeevani or for a recheck, then follows up until it's done.

Doctor approves

Every plan that changes care waits for the centre's doctor, using ADK tool confirmation. Approvals can be done in batches, in seconds.

Counsellor discloses

Infection results never appear in any agent message. The agent only books "a confidential conversation", and follows up afterwards.

Red flags skip the AI

BP $\geq 180/120$, or symptoms such as chest pain or breathlessness, trigger fixed urgent templates (ICMR 2026). No model judgement is involved.

It never diagnoses or prescribes (Telemedicine Practice Guidelines §5.4). It never offers anything for donating (D&C Rule 122EA). Care never depends on donating.

Gen AI does the navigator's work. Ordinary code does the safety work.

INPUTS

Donor on WhatsApp or web

text, photo, voice (Hindi, Kannada, English)

Blood-centre register

photo or export of each donation's readings

The donor's own reports

company or insurer check-up, any lab format

PEOPLE IN CHARGE

Doctor console

approve, edit or reject plans, in batches

Counsellor console

disclosure queue; the only role that sees flags

CLOUD RUN · ADK AGENTS

Record builder · Gemini 3.8 Flash

image or PDF → readings with a confidence per field → FHIR

Finding triage · fixed rules

ICMR hypertension and diabetes workflows, haemoglobin; red flags go to templates

Navigator · Gemini 3.8 Flash (Live for voice)

explains, plans, drafts the doctor's summary, follows up, checks back at the next donation

Tools

read_record · find_facility (Maps) ·

book_slot

(needs the doctor's confirmation) · schedule_followup · request_counselling · share_summary

DATA AND ANALYTICS

Cloud Healthcare API · FHIR store

Observations, DiagnosticReports, ready for ABHA

Firestore

conversations, plans, audit; counsellor flags in a restricted collection

BigQuery

findings-to-care funnel and evaluation

ROUTES TO EXISTING CARE

Ayushman Arogya Mandir · eSanjeevani · NVHCP · ABHA

free public care and records the state already pays for

From a dead-end deferral to care, then back to donating

Evening · Hindi

"Your BP was 148/94 today. It has gone up at each of your last 4 donations. High BP usually has no symptoms, which is why it matters."

He sends a photo

Last year's company check-up. Gemini reads **HbA1c 6.1%** (prediabetes range, ICMR) and adds his father's history.

Doctor · 15 s

Dr Rao sees the trend, the report extract and the cited protocol, and approves the plan.

Booked

A free BP and sugar check at the Ayushman Arogya Mandir near his home, Saturday 10 am. The summary is shared to his records.

3 months later

He's under care, the doctor clears him, and he's welcomed back. **He donates again.**

Counsellor console

"14 donors need a confidential conversation. The agent has reached 11 and booked them in. **You have 3 left to call.**"

The agent never states a result. It books the conversation and follows up afterwards (synthetic data).

Red flag, no AI

Donor D-017 has BP 184/118: an urgent template goes out in under 1 second, and the doctor is alerted. If symptoms are mentioned, the template points to emergency care (112).

From a third of findings followed up to every finding followed through

MEASURE	TODAY	WITH TRACEDROP	SHOWN IN
Findings with a completed next step	~1 in 3 (reactive); ~0 (deferrals)	Target ≥2 in 3	Pilot
Time from finding to a booked next step	Weeks, or never	Same day	Prototype
Calls left for one counsellor (synthetic queue of 14)	14	3	Prototype
Donors who leave knowing what their numbers mean and what to do	~None	All	Prototype

Prototype evaluation (fill in on 15 Oct from eval/results.json)

[]%

field accuracy on [N] real Indian lab reports (the team's own, with consent)

0 / 50

target: infection-result leaks in adversarial tests

100%

target: red flags handled by fixed rules with no AI, as unit-tested

[]/5

Hindi and Kannada quality, rated by native speakers

It routes to care the state already pays for. No new tests, no new budget.

Who benefits

- **Donors:** silent conditions found and followed up
- **Women:** 57% are anaemic, and they're the most often deferred
- **People with hepatitis:** curable, free treatment under NVHCP
- **Blood centres:** they meet their counselling duties, and cleared donors come back
- **Patients:** more regular donors

Who pays (donors never do)

- **Blood centres and SBTCs:** counselling and NVHCP linkage duties
- **Employers:** annual check-ups under the OSH Code (in force Nov 2025)
- **Insurers:** wellness programmes allowed by IRDAI, with consent

Path to scale

- 3 Bengaluru centres → every finding type → every donor
- SBTCs and NHM; ABHA and eSanjeevani connection
- JAPAC: Japan returns results and Taiwan screens, but neither navigates donors to care

Built to be trusted

Synthetic data in the prototype · thresholds from ICMR 2026 and WHO · the doctor approves and the counsellor discloses · DPDP consent, and employers see aggregates only · a deployment would register as CDSCO medical device software and follow ICMR's 2023 AI ethics guidelines.



Every donation is a health check. TraceDrop makes sure it counts.

Today, India's blood centres promise donors a health check every time they give, about 1.5 crore times a year. What the check finds is rarely followed up. With Gemini, we remove that limit and deliver the same promise 10x better.